



## ***Playful Minds***

Parental playfulness within the relation of pretend play and executive functioning

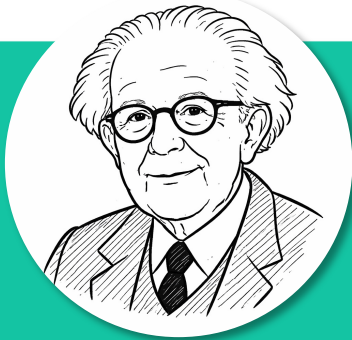


**Tobias Constien**  
UCD Babylab, August 27<sup>th</sup>, 2026

# What is Play?

*"Play is a very 'fuzzy' concept."*

(Lillard 2013, p. 426)



Jean Piaget



Lev Vygotsky



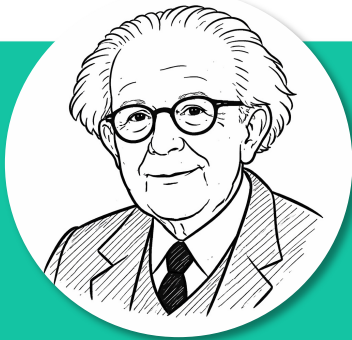
Maria Montessori



# What is Play?

*“When you’re having fun doing what you’re doing and you like to do it.”*

(Lucy, cited in Miller et al. 2008, p. 410)



Jean Piaget



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# What is Play?

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(Lucy, cited in Miller et al. 2008, p. 410)

## Continuum of Play

Not Playful



Fully Playful





# What is Play?

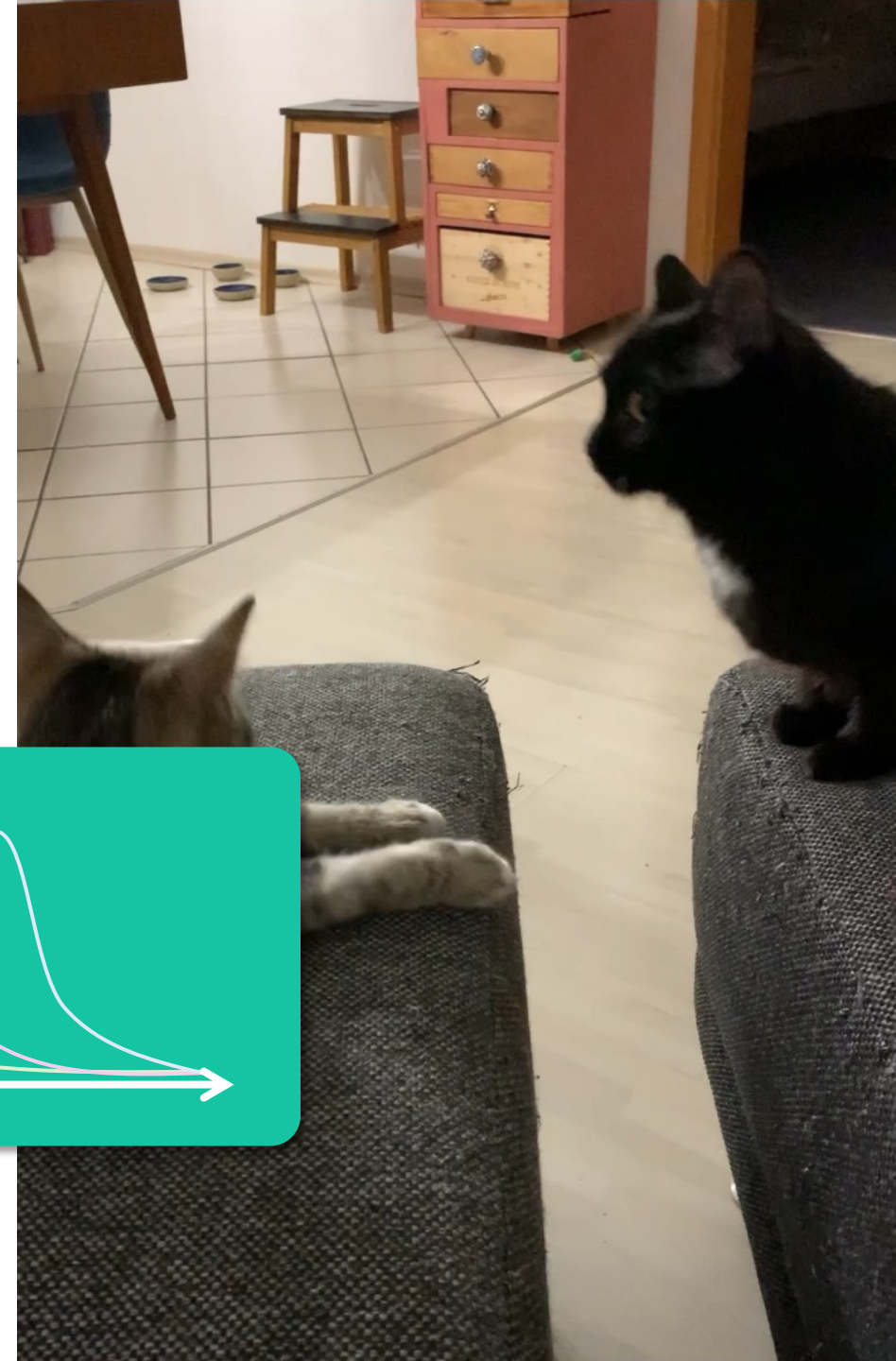
*“When you’re having fun doing what you’re doing and you like to do it.”*

(Lucy, cited in Miller et al. 2008, p. 410)

**Varies categories of play exist.**

Sensorimotor play, physical play, rough-and-tumble play, etc.

**Prevalence** varies across age.



- Involves imagination, creativity, and fantasy.
  - Object substitution
  - Object attribution
  - Role play
- Emerges around 18 months of age and peaks during the preschool years.
- Related to the development of social competence, language, and executive functions.

Constien et al., 2026; Li & Zhang, 2026; Quinn et al., 2018; Smits-van der Nat et al., 2024

# Pretend Play







# Linking Pretend Play and EF

**Executive  
Functions**

**Representational  
Competence**

**Pretend  
Play**

**Practice**

**Parental Playfulness**





# Adults play too!



Relationship  
satisfaction



Physical  
Activity



Mental  
Health



Parenting

# The Current Study

## *Hypothesis 1*

Toddlers' ability to engage in pretend play will positively predict toddlers' executive functions.

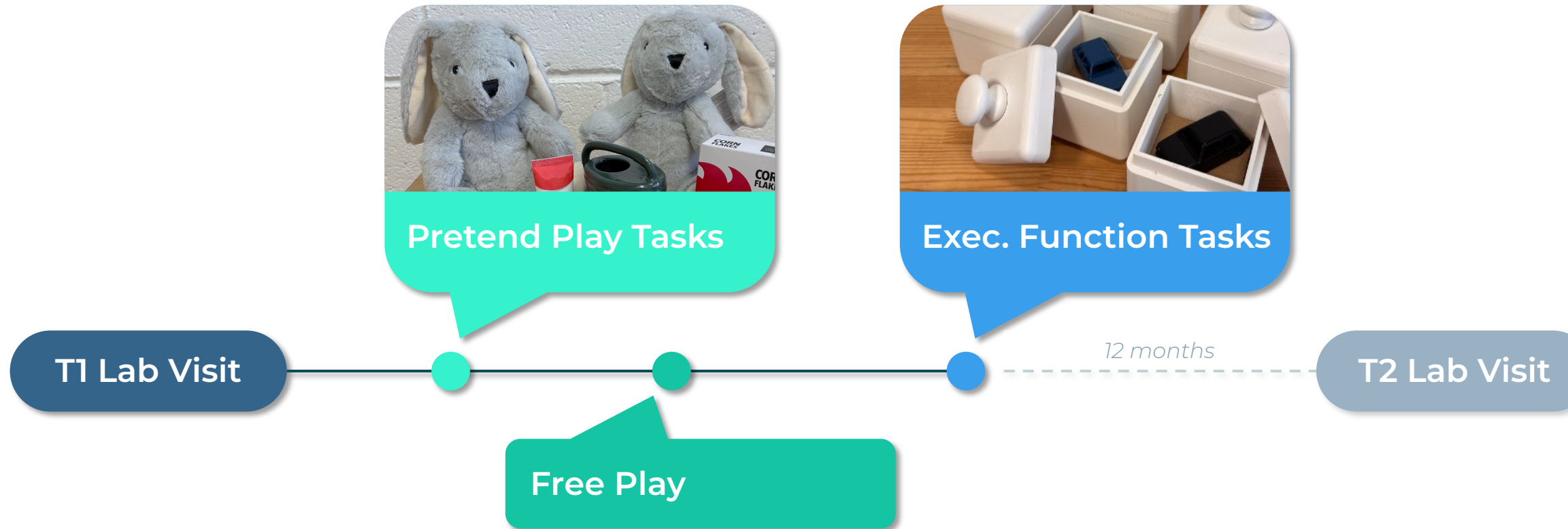
## *Hypothesis 2*

Parental playfulness will be positively associated with toddlers' ability to engage in pretend play.

*pre-registered on OSF*



# The Current Study







# Participants

## ● 100 Toddlers

- $M = 23.51$  months ( $SD = 2.75$ )
- Range: 20 – 28 months



- 86% White Irish, 8% other White, 6% other ethnic background

## ● 100 Parents

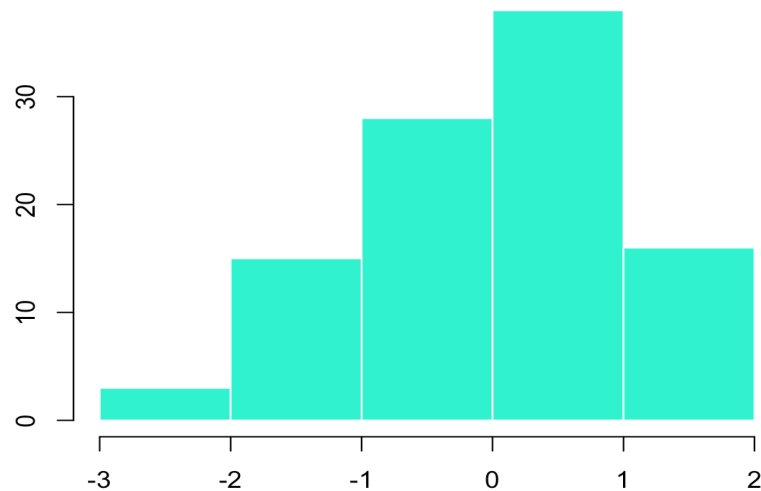
- 90% Mothers, 10% Fathers
- 93% third-level education

# Measures

## ● Pretend Play

- Aggregated Score across three tasks:

Pretend Play Comprehension | Elicited Pretend Play Actions | Impersonation Interview



*Standardised Score*

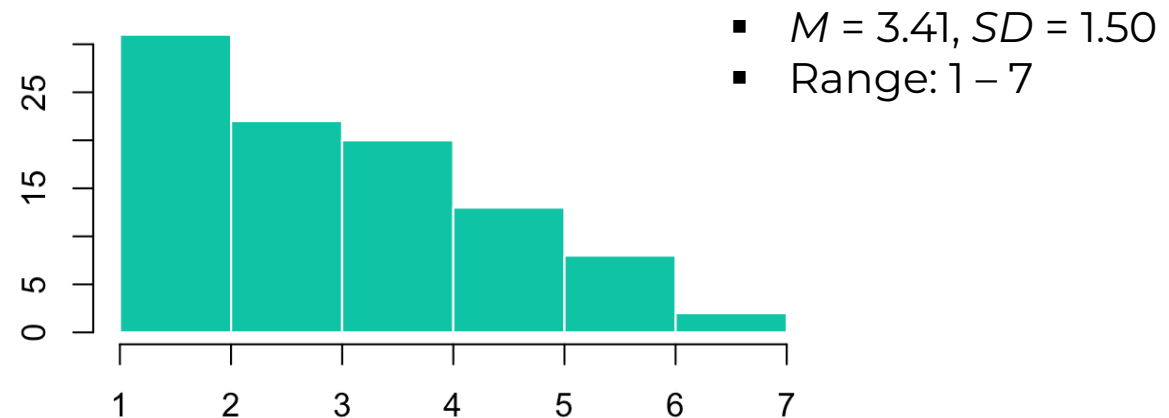
- $M = 0, SD = 1$
- Range: -2.28–1.66



# Measures

## ● Parental Playfulness

- The extent to which a parent shows imagination, creativity, and enjoyment in play with their child.
- Global score using an observational coding scheme ( $ICC = .77$ ).

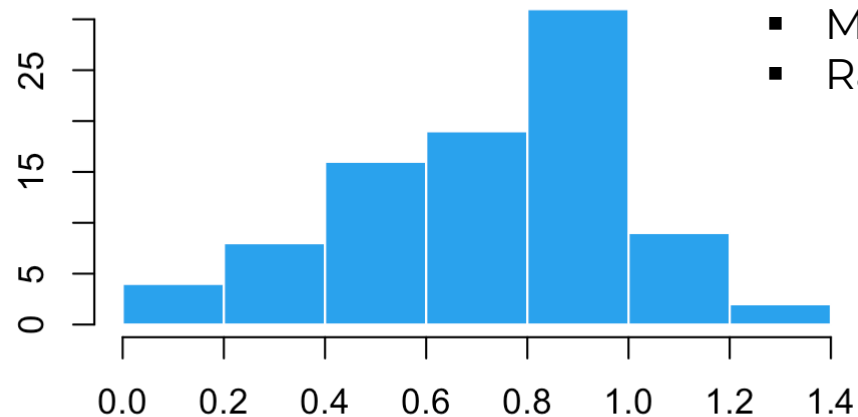




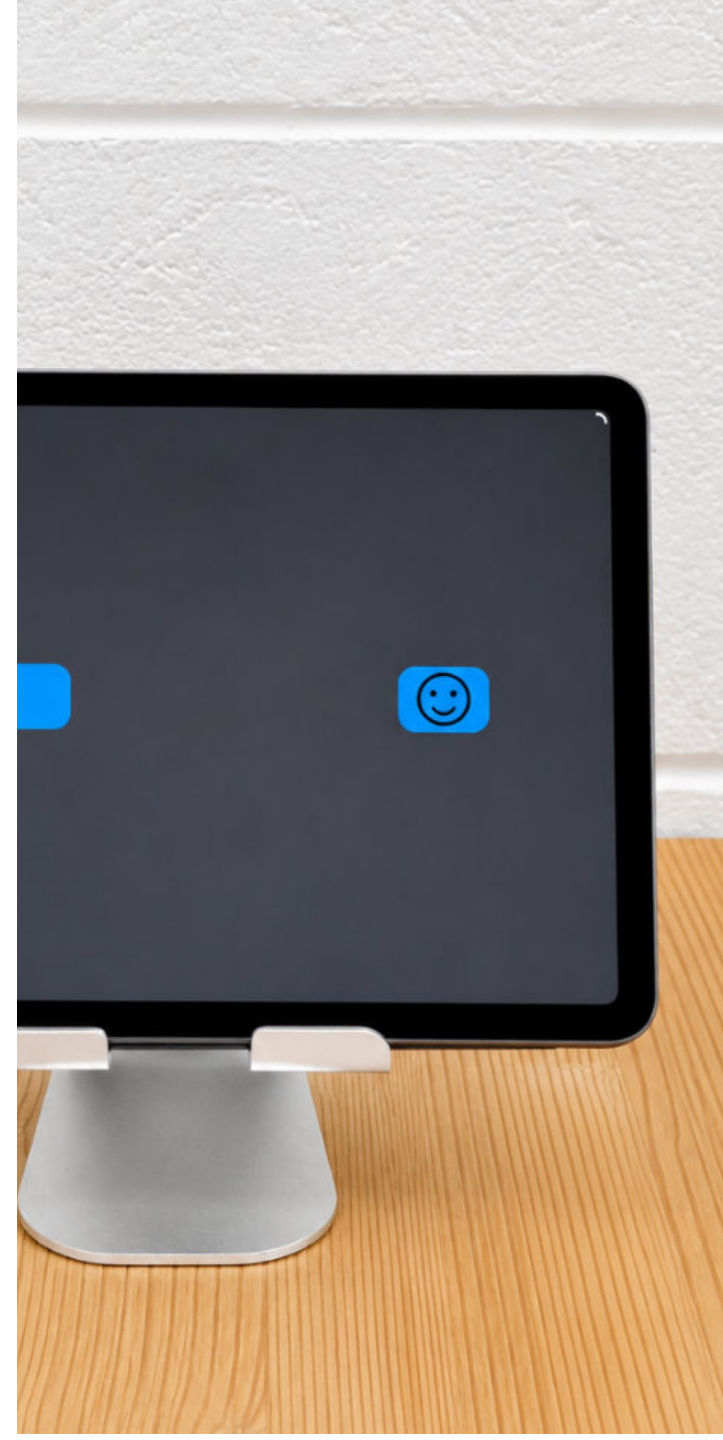
# Measures

## ● Early Childhood Inhibitory Touchscreen Task

- Used to assess response inhibitory control.
- Accuracy score on inhibitory trials (adjusted for prepotent accuracy)



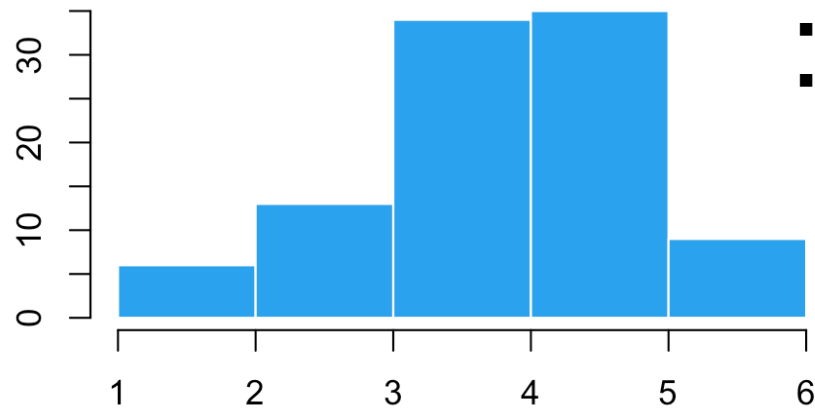
- $M = 0.74$ ,  $SD = 0.28$
- Range: 0 – 1.33



# Measures

## ● Six Boxes Task

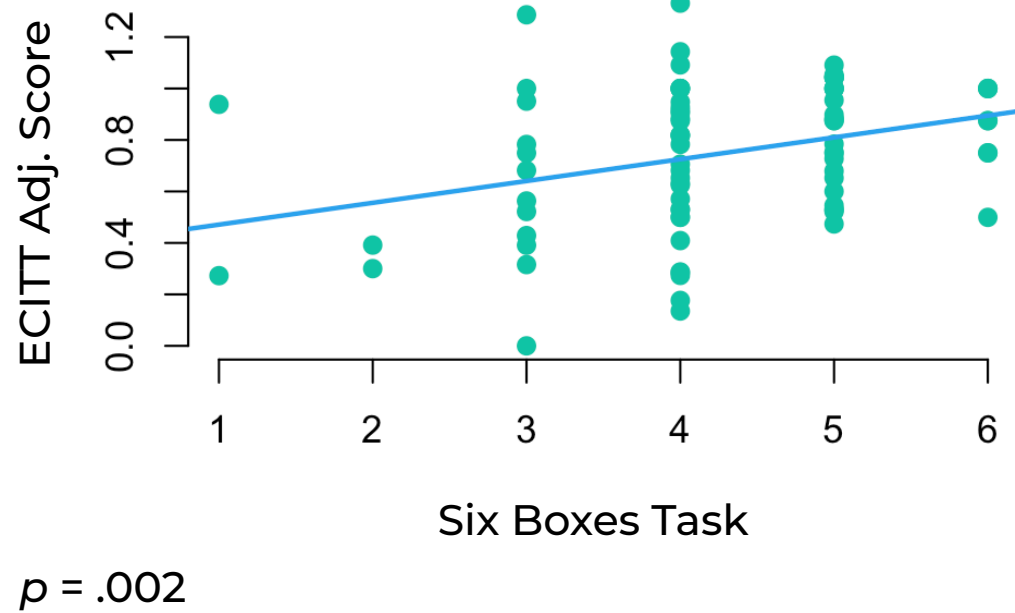
- Used to assess working memory.
- Scored by number of retrieved cars after six trials.



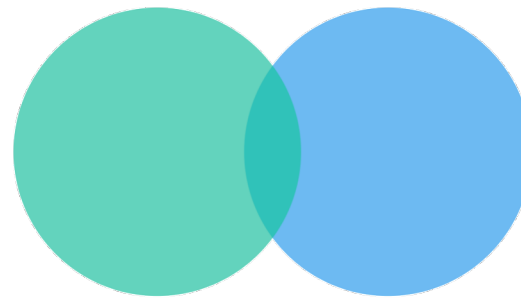
- $M = 4.26, SD = 1.1$
- Range: 1 – 6 Cars



# Measures



**Correlation:  $r = .32$**   
Shared Variance: 10%



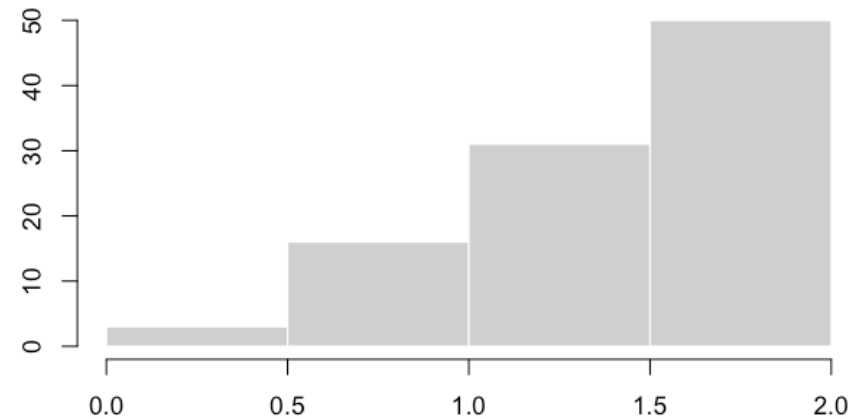




# Measures

## ● Language Skills

- 5-15 Toddler Questionnaire – Language Subscale
  - My child has difficulty understanding words.
  - My child has difficulty saying single words and short sentences.

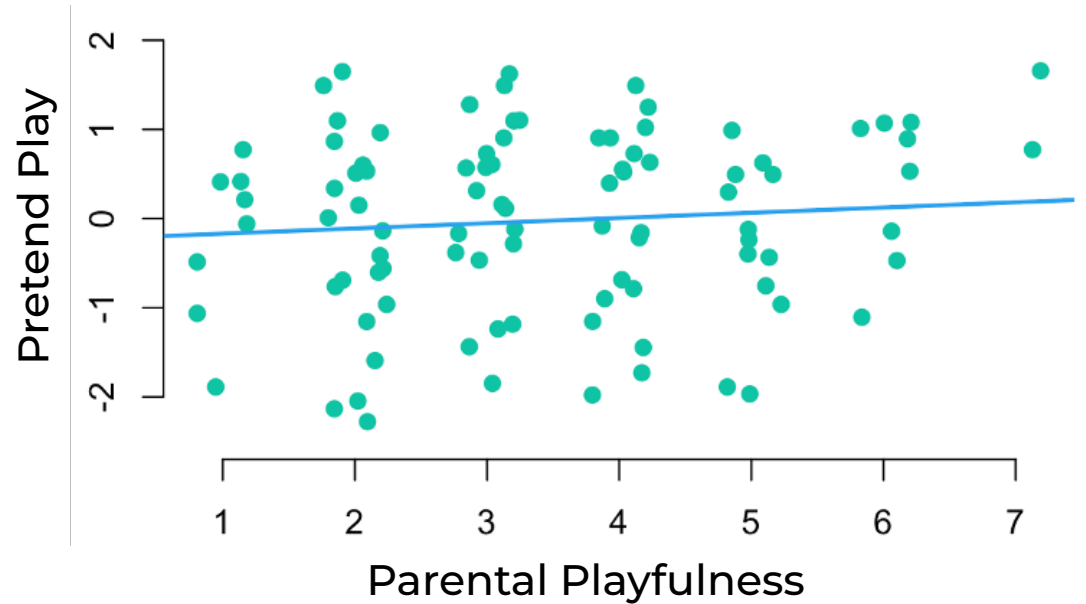


- $M = 1.44, SD = 0.41$
- Range: 0 - 2



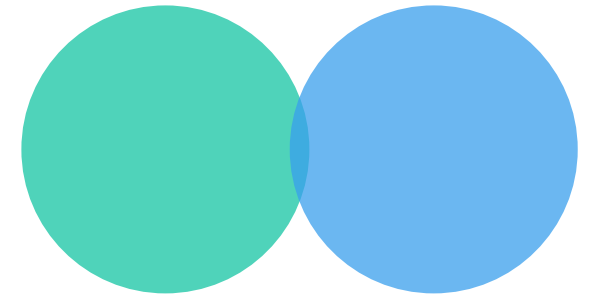
# Pretend Play and Parental Playfulness

*Two unrelated constructs?*



**Correlation:  $r = .13$**

Shared Variance: 2%

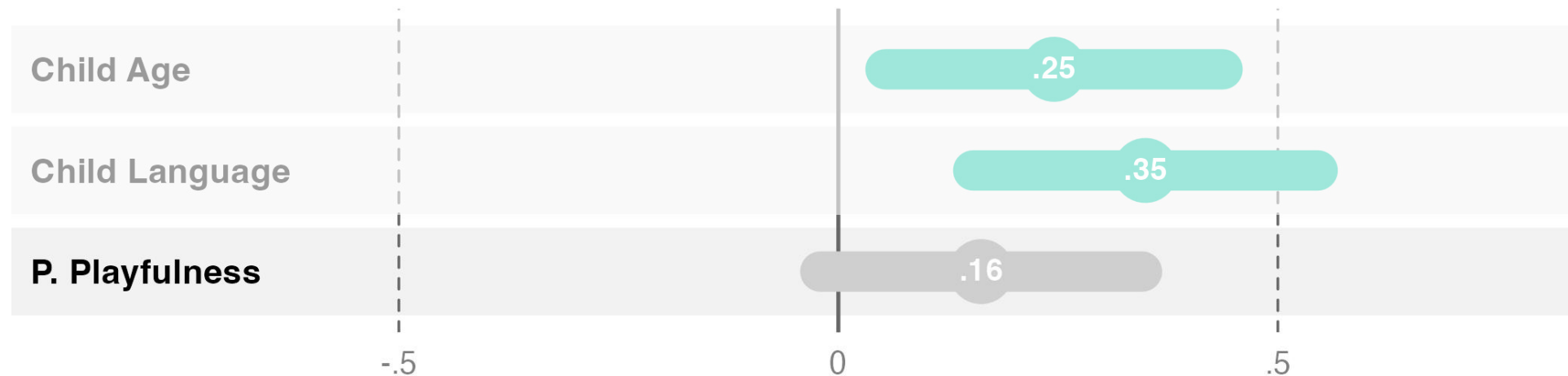


# Pretend Play and Parental Playfulness

*Two unrelated constructs?*

## Predicting Toddlers' Pretend Play Ability

$F(4, 92) = 10.38, p < .001, \text{Adj. } R^2 = .23$



$N = 96$ . Standardised coefficients and their 95% confidence interval are shown.

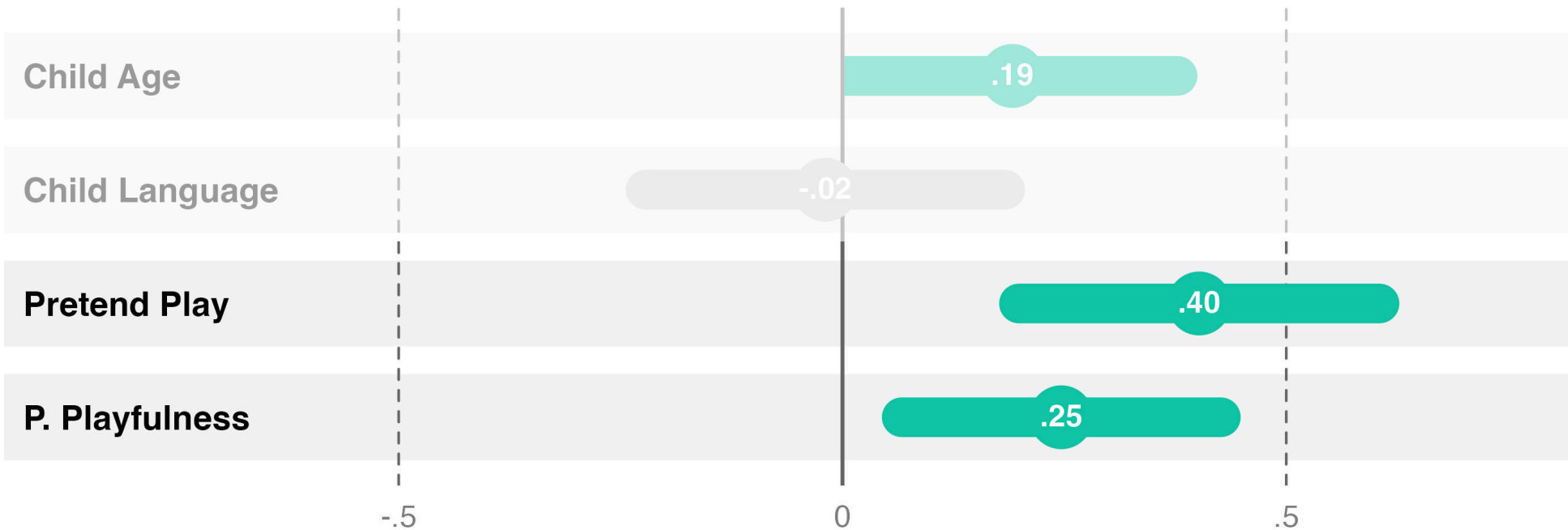


# Play and Executive Functions

*Unique contributions of parents and toddlers play*

## Predicting Toddlers' Working Memory

$F(4, 88) = 10.82, p < .001, \text{Adj. } R^2 = .30$



$N = 93$ . Standardised coefficients and their 95% confidence interval are shown.

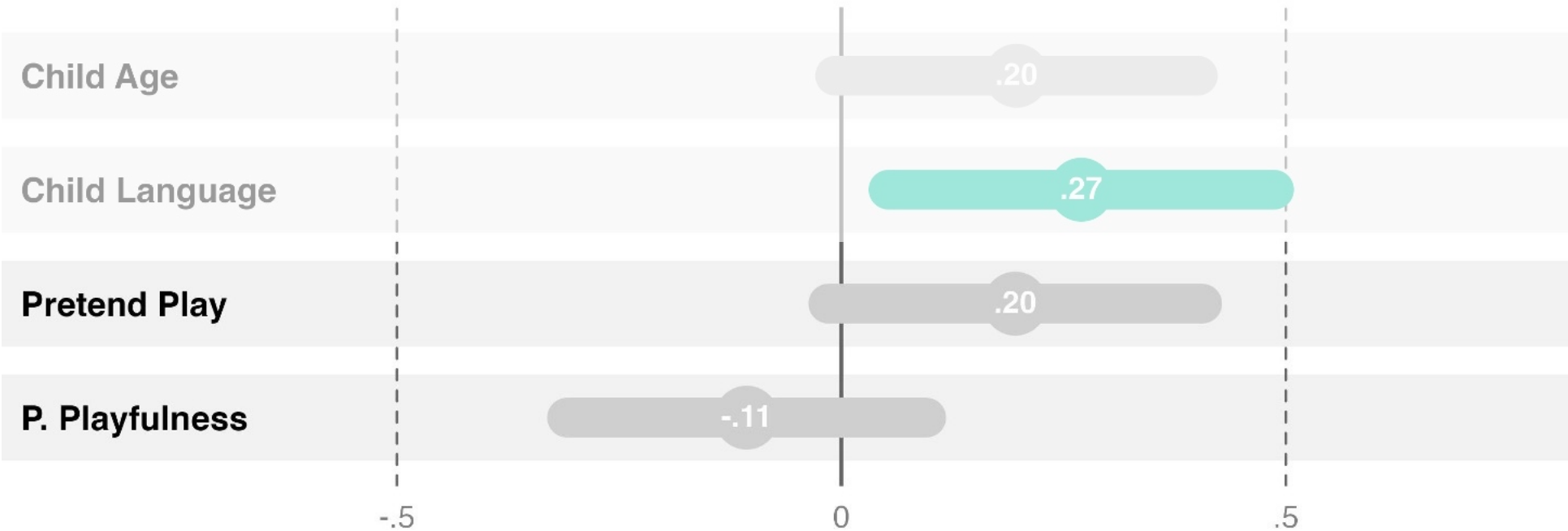


# Play and Executive Functions

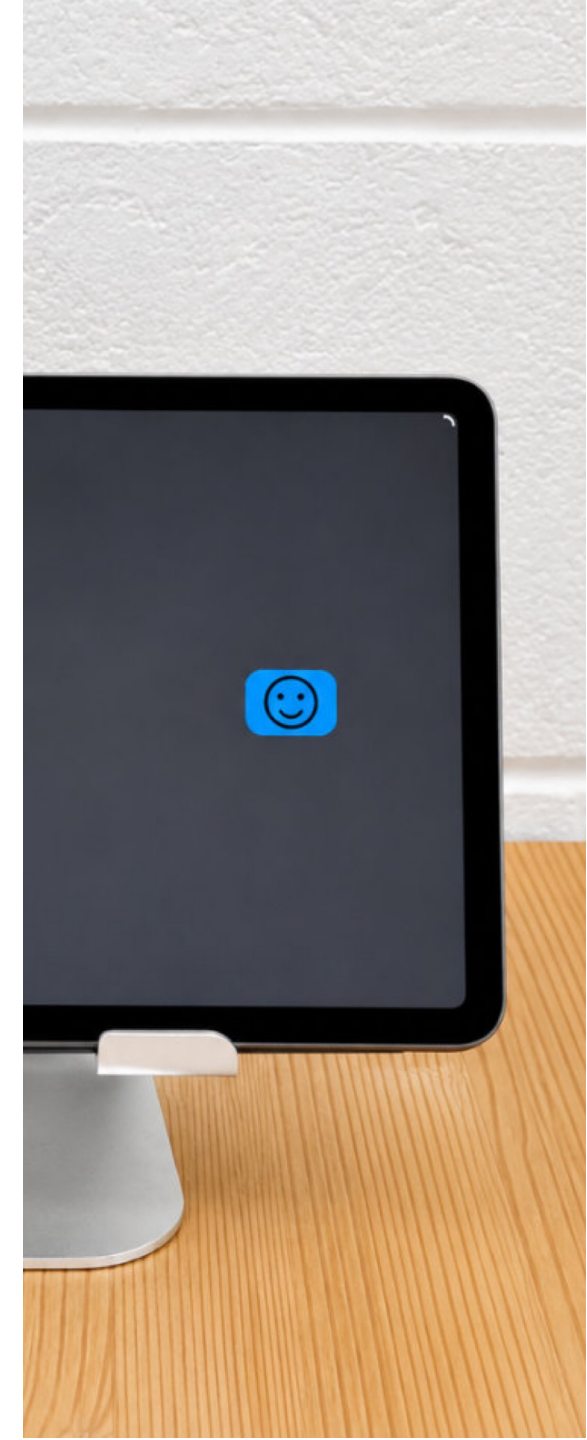
*Unique contributions of parents and toddlers play*

## Predicting Toddlers' Response Inhibition

$F(4, 80) = 7.11, p < .001, \text{Adj. } R^2 = .23$



$N = 85$ . Standardised coefficients and their 95% confidence interval are shown.



# Take Aways

## ● Not *just* play

Executive functions are shaped through **interactions**, specifically in instances where children are required to **maintain representations**.

- Current findings reflect previous meta-analytic results and theoretical frameworks.
- Unique associations to working memory but not inhibitory control may indicate age effects.
- Longitudinal stability of relations requires further study.





# Take Aways

## ● Being **silly** pays off

Parental playfulness has broader implications for children's development than shaping children's play.

- Playful interactions create manageable but challenging cognitive demands (i.e., *zone of proximal development*).
- Reflects previous findings regarding developmental benefit of parental playfulness.





# Ways to be *playful*

- Build a tower with stuff you have lying around on your desk.
- Make up the cast of Harry Potter only using your friends.
- Challenge a friend to a round of Tic-Tac-Toe.
- Take a different way home.
- Write a two-line poem.

Thank you for your attention,  
at this Psychology convention!



# Stay in touch!

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*Supervisor, University College Dublin*



## Download these slides!



[tinyurl.com/playful-minds](https://tinyurl.com/playful-minds)

*“It’s easy to fly: Just bend  
your knees and jump!”*

(Child 20, cited in Ramsden et al. 2026, p. 12)

